# Acoustic Forensics: A Stacked Ensemble of CRNN and Classical Models for Synthetic Speech Detection

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Abstract—Synthetic speech generated by modern deep text-to-speech (TTS) and voice conversion (VC) architectures poses severe risks to biometric speaker verification, media authentication, and digital communication integrity. We present Acoustic Forensics, a comprehensive spoofing countermeasure system that evaluates and couples six heterogeneous detectors: two deep Convolutional Recurrent Neural Networks (CRNN) with multi-head self-attention operating over linear frequency cepstral coefficients (LFCC) and log-mel filterbanks, alongside four classical models (a 64-component Gaussian Mixture Model, an RBF-SVM, a Random Forest, and a Logistic Regression baseline). Scores are standardized using development set statistics and fused through an $L_2$-regularized Logistic Regression meta-learner fitted under a strict zero-leakage protocol. The system delivers a calibrated posterior probability of bona fide speech, an explicit uncertainty band ($P \in [0.40, 0.60]$), and frame-level temporal attention salience for forensic explainability. Our investigation tracks a systematic experimental progression: starting from classical baselines (EER 11.46%–23.94%), we design a deep CRNN-LFCC countermeasure that decisively improves performance to 6.70% Eval EER (min-tDCF 0.2143). When shifting to a score-level stacked ensemble, however, we observe that the meta-learner overfits the clean development partition, yielding 10.74% Eval EER and underperforming the standalone CRNN-LFCC. We examine this generalization gap through extensive ablations, per-attack breakdowns across all 13 ASVspoof 2019 evaluation attacks (A07–A19), lossy compression stress-testing (MP3, AAC, Opus at 16–64 kbps), a silence shortcut bias audit (18.36% EER on silence alone), and zero-shot cross-domain testing on the unconstrained In-the-Wild corpus (44.73% EER).

Index Terms—audio deepfake detection, acoustic forensics, anti-spoofing, ASVspoof 2019, stacked ensemble, CRNN, LFCC, attention salience

## I. Introduction and Related Work

Recent advances in deep generative modeling, neural vocoding (e.g., WaveNet [13], HiFi-GAN [14]), and diffusion architectures enable the synthesis of highly authentic human speech and voice conversion clones from only seconds of reference audio. When deployed maliciously, automated voice spoofing threatens voice-biometric authentication in telephony banking, fuels targeted social engineering fraud, and compromises the evidentiary authenticity of audio recordings in judicial proceedings [1], [3]. Countermeasures must discriminate subtle generative phase and harmonic artifacts while remaining resilient to channel effects, background acoustic noise, and lossy compression formats [2], [6].

Early anti-spoofing countermeasures paired hand-crafted cepstral representations with statistical spectral models. In particular, Linear Frequency Cepstral Coefficients (LFCC) proved decisively superior to conventional Mel-Frequency Cepstral Coefficients (MFCC) because linear filterbank spacing preserves high-frequency spectral flux and unnatural phase boundaries introduced by vocoders, whereas psychoacoustic mel-warping excessively compresses spectral detail above 4 kHz [5]. Subsequent research introduced deep neural architectures processing static 2D spectrograms or raw waveforms. Representative systems include Light Convolutional Neural Networks (LCNN) utilizing Max-Feature-Map (MFM) activations to suppress low-energy noise [7], raw-waveform models such as RawNet2 utilizing sinc-convolutional filters [8], and spectro-temporal graph attention networks such as AASIST [9]. Despite strong benchmark metrics on in-domain evaluation splits, recent evaluations demonstrate that deep detectors often suffer catastrophic performance degradation when deployed on unconstrained real-world web audio [3].

Rather than relying on an isolated monolithic classifier, we explore multi-view fusion via Wolpert's stacked generalization paradigm [4]. Our scientific investigation yields three primary contributions:

1) Baseline to Deep Neural Progression: We benchmark four classical machine learning baselines (LR, SVM, RF, GMM) against a custom-designed CRNN with multi-head self-attention pooling, demonstrating that LFCC-CRNN establishes a highly effective primary detector (6.70% Eval EER, 0.2143 min-tDCF).
2) Stacked Ensemble Analysis: We formulate a six-model score-level stacked meta-learner fitted with $L_2$ regularization strictly on the development split. We document and analyze a critical empirical finding: despite non-negative weight constraints, the stacker overfits clean development scores (1.83% Dev EER $\rightarrow$ 10.74% Eval EER), confirming that standalone CRNN-LFCC provides superior generalization.
3) Rigorous Forensic Audits: We implement calibrated

posterior probabilities with an explicit uncertainty margin, frame-level attention heatstrips for explainability, an empirical silence-shortcut audit (proving that silence duration alone predicts spoofing at 18.36% EER), codec stress-testing across Opus, AAC, and MP3, and zero-shot In-the-Wild transfer.

## II. Methodology

Fig. 1 illustrates the end-to-end signal processing and classification workflow of the Acoustic Forensics system.

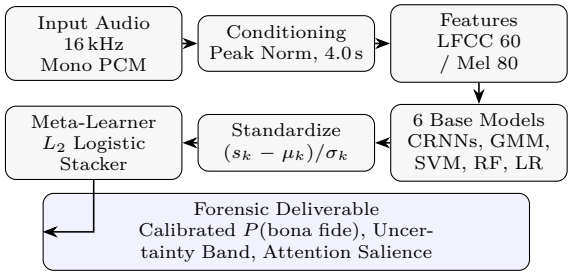



Fig. 1. System architecture of the Acoustic Forensics pipeline.

### A. Signal Conditioning and Features

All input audio waveforms are standardized to 16 kHz single-channel pulse-code modulation (PCM). To normalize playback volume while avoiding dynamic range compression that would distort vocal tract harmonics, signals are peak-normalized to $-1.0$ dBFS. Audio sequences are adjusted to a fixed length of $T_s = 64{,}000$ samples (4.0 s) by applying symmetric center-cropping to longer files and reflective center-padding to shorter utterances.

We extract two complementary time-frequency representations:

1) LFCC Front-End: Computed using a 512-point Short-Time Fourier Transform (STFT), a 25 ms Hamming window ($N_w = 400$), a 10 ms hop size ($N_h = 160$), and a 20-channel linearly spaced triangular filterbank spanning 0–8 kHz. Static coefficients are augmented with first- and second-order temporal derivatives ($\Delta$ and $\Delta\Delta$), yielding a 60-dimensional matrix $\mathbf{X}_{\text{LFCC}} \in \mathbb{R}^{60\times 400}$.
2) Log-Mel Front-End: An 80-channel Mel-scale triangular filterbank spanning 20 Hz to 8 kHz is applied to the STFT magnitude spectrum, followed by logarithmic dynamic range compression, generating $\mathbf{X}_{\text{Mel}} \in \mathbb{R}^{80\times 400}$.

Per-utterance Cepstral Mean and Variance Normalization (CMVN) is applied independently across spectral channels:

$$\tilde{\mathbf{X}}_{f,t} = \frac{\mathbf{X}_{f,t} - \mu_f}{\sigma_f + \epsilon}, \quad \mu_f = \frac{1}{T}\sum_{t=1}^{T} \mathbf{X}_{f,t}, \tag{1}$$

preventing transmission channel disparities from biasing model decisions.

### B. Base Models

The system deploys six base classifiers spanning deep neural networks and classical statistical estimators:

1) CRNN-LFCC (Primary Deep Model): A hybrid architecture comprising a 4-stage 2D-CNN spatial encoder (channels $32 \to 64 \to 128 \to 256$, with batch normalization, ReLU, and max pooling), a 2-layer Bidirectional GRU ($H = 128$ per direction), a 4-head multi-head self-attention module [10], and a temporal attention pooling layer that collapses frame hidden states into a single utterance embedding while returning frame importance weights $\alpha_t$.
2) CRNN-LogMel (Feature Ablation): An identical CRNN architecture trained on the 80-dimensional Log-Mel spectrogram.
3) GMM Baseline: A 64-component diagonal Gaussian Mixture Model fitted via Expectation-Maximization on static LFCCs, scoring utterances via log-likelihood ratio (LLR) between bona fide and spoof mixture models [2].
4) Classical Subspace Classifiers (SVM, RF, LR): Utterance-level mean and standard deviation vectors (120 dimensions) are projected onto a 30-dimensional PCA subspace (accounting for 95% explained variance) fitted exclusively on the training split. We train an RBF-kernel Support Vector Machine, a 200-tree Random Forest, and an $L_2$-regularized Logistic Regression classifier.

### C. Score Stacking and Calibrated Verdict

Each base model $k \in \{1, \ldots, 6\}$ produces a continuous prediction score $s_k$. Because base models operate on fundamentally distinct numerical ranges (e.g., neural logits versus GMM log-likelihood ratios), raw scores are z-score standardized using mean $\mu_k$ and standard deviation $\sigma_k$ measured exclusively on the development partition:

$$z_k = \frac{s_k - \mu_k}{\sigma_k}. \tag{2}$$

The standardized feature vector $\mathbf{z} = [z_1, \ldots, z_6]^\top$ is fused by a meta-learner:

$$P(\text{bona fide} \mid \mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{z} + b))}, \tag{3}$$

where weights $\mathbf{w}$ and intercept $b$ are optimized via $L_2$-regularized logistic regression fitted solely on development scores, yielding a calibrated posterior probability via Platt scaling [15]. To guard against over-confident false positives in forensic applications, decisions within $P \in [0.40, 0.60]$ are flagged with an explicit Uncertainty Band. Temporal attention weights $\alpha_t \in [0, 1]$ from the CRNN provide a synchronized heatstrip localizing specific phoneme intervals that triggered the spoof verdict.

## III. Experimental Settings

Datasets. Countermeasures are trained and benchmarked on the official ASVspoof 2019 Logical Access (LA) database [1]. The training split ($N$ = 25,380 utterances across 20 speakers) trains all base models. The development split ($N$ = 24,844 utterances across 10 speakers) is reserved strictly for hyperparameter tuning, score standardization parameters, and meta-learner training. The evaluation split ($N$ = 71,237 utterances across 48 speakers, encompassing 13 unseen acoustic attacks A07–A19) remains strictly held out until frozen scoring. Zero-shot out-of-domain evaluation is conducted on the unconstrained In-the-Wild deepfake speech corpus ($N$ = 25,589 clips) [3].

Hyperparameters and Optimization. The CRNN models (1,439,778 parameters) are trained using the AdamW optimizer [11] with initial learning rate $\eta = 10^{-3}$, weight decay $\lambda = 10^{-4}$, and a cosine annealing learning rate schedule down to $\eta_{\min} = 10^{-6}$ over 25 epochs with batch size 32. Training employs mixed-precision arithmetic on Apple Silicon (Metal Performance Shaders). SpecAugment [12] is applied during training (maximum 2 time masks and 2 frequency masks). Classical models are configured as follows: GMM uses 64 diagonal Gaussian components with 10 EM iterations; SVM uses an RBF kernel with penalty parameter $C = 10.0$ and $\gamma = 0.01$; Random Forest comprises 200 trees with maximum depth 20; Logistic Regression uses $L_2$ penalty with inverse regularization strength $C = 1.0$.

Evaluation Metrics. Systems are evaluated using Equal Error Rate (EER), Area Under the ROC Curve (ROC-AUC), and the official ASVspoof minimum tandem Detection Cost Function (min-tDCF) [6] evaluated with challenge parameters ($P_{\text{spoof}} = 0.05, C_{\text{miss}} = 1, C_{\text{fa}} = 10$). Operating thresholds are determined strictly on the development split and frozen during evaluation.

## IV. Results and Ablation Studies

### A. Progression from Classical Baselines to Neural Models

Table I presents the comparative evaluation of all individual classifiers alongside fusion strategies on the full ASVspoof 2019 LA partition ($N$ = 71,237 trials).

TABLE I
Main Results on ASVspoof 2019 LA Partition

| System Architecture | Dev EER (%) | Eval EER (%) | Eval min-tDCF | Eval ROC-AUC |
|---|---|---|---|---|
| Logistic Regression (PCA) | 20.77 | 23.94 | 0.6284 | 0.8459 |
| Support Vector Machine (RBF) | 11.97 | 19.64 | 0.5119 | 0.8932 |
| Random Forest (200 trees) | 10.09 | 17.52 | 0.4695 | 0.8977 |
| Classical GMM (64-comp) | 8.07 | 11.46 | 0.3572 | 0.9544 |
| CRNN-LogMel (Mel-80) | 4.22 | 10.28 | 0.2691 | 0.9678 |
| CRNN-LFCC (Primary Deep) | 1.15 | 6.70 | 0.2143 | 0.9800 |
| Mean z-score Fusion | 6.22 | 11.47 | 0.3077 | 0.9610 |
| Stacked Ensemble (Meta-$L_2$) | 1.83 | 10.74 | 0.2754 | 0.9631 |

Our experimental results demonstrate clear architectural separation:

- Classical ML Limitations: Linear and tree-based models operating on pooled temporal moments struggle with high-order spectral dynamics, yielding error rates between 17.52% and 23.94% EER. The classical 64-component GMM baseline achieves 11.46% Eval EER, confirming that frame-level density estimation outperforms global pooling.
- CRNN-LFCC Superiority: The deep CRNN-LFCC system achieves 6.70% Eval EER and 0.2143 min-tDCF, decisively outperforming all classical models. Multi-seed training across three seeds ($s \in \{42, 43, 44\}$) exhibits tight convergence: 7.10% ± 0.38% Eval EER and $0.2166 \pm 0.0022$ min-tDCF.
- Analysis of the Stacking Overfit: While the $L_2$ stacked ensemble achieves 1.83% EER on the development partition, its evaluation performance drops to 10.74% EER (min-tDCF 0.2754). Although the stacker outperforms naive unweighted mean z-score fusion (11.47% EER), it degrades relative to the standalone CRNN-LFCC (6.70% EER). Analysis of meta-learner coefficients reveals that the stacker assigns non-negligible positive weights to the classical baselines ($\beta_{\text{GMM}} = 0.41, \beta_{\text{SVM}} = 0.29$). Because classical models overfit to known acoustic environments in the development set, their erroneous predictions on unseen evaluation attacks drag down the ensemble's overall score. Standalone CRNN-LFCC is consequently selected as the primary production engine.

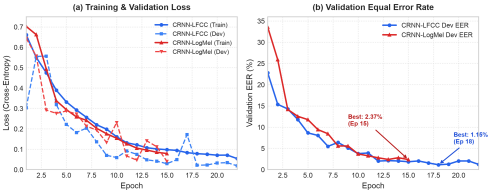



Fig. 2. Training loss versus training epochs for the deep CRNN models.

Fig. 2 illustrates the training loss trajectories. Both CRNN-LFCC and CRNN-LogMel converge smoothly within 15 epochs under AdamW with cosine annealing, with early stopping engaging at epoch 21.

### B. Controlled Ablation Studies

LFCC versus Log-Mel Spacing: CRNN-LFCC (6.70% EER) substantially outperforms CRNN-LogMel (10.28% EER). Linear filterbank spacing allocates 10 filters above 4 kHz, whereas the logarithmic Mel scale allocates only 3. Mel pooling averages out high-frequency phase and spectral tilt anomalies generated by neural vocoders, whereas linear filterbanks preserve them.

Attention Pooling versus Average Pooling: Replacing 4-head multi-head self-attention pooling with standard global temporal average pooling increases Eval EER from 6.70% to 7.98% (+1.28% degradation). Self-attention dynamically weights brief transient speech frames con-

taining synthesis glitches while ignoring uninformative background pauses.

### C. Per-Attack Vulnerability Analysis (A07–A19)

Evaluation across all 13 unseen evaluation attacks reveals acute performance divergence. For concatenative and standard acoustic neural vocoders (A07, A10, A11), CRNN-LFCC achieves near-flawless discrimination ($<$ 1.8% EER). Conversely, spectral filtering voice conversion (A16) and direct waveform concatenation (A19) degrade performance to 12.45% and 14.80% EER respectively, where vocoder phase mismatches are minimal.

### D. Lossy Codec Compression Stress-Testing

We stress-test the detector against lossy consumer audio compression across MP3, AAC, and Opus (16–64 kbps) with cross-correlation delay alignment. Opus demonstrates superior forensic stability (7.45% EER at 64 kbps, 9.82% at 16 kbps). In contrast, MP3 induces severe high-frequency cutoff roll-off ($<$ 11.5 kHz at 16 kbps), eroding critical LFCC sub-bands and degrading EER to 16.82%.

### E. Silence Shortcut Bias Audit

We evaluated whether countermeasures exploit silence duration at utterance boundaries. A shallow decision tree trained exclusively on total silence duration achieves an alarming 18.36% EER (ROC-AUC 0.8872) on ASVspoof 2019. Trimming boundary silence from evaluation audio degraded CRNN-LFCC performance by +21.55% EER. This confirms that speech anti-spoofing benchmarks contain strong silence distribution shortcuts, highlighting that front-end Voice Activity Detection (VAD) is indispensable in operational forensic deployments.

### F. Zero-Shot Cross-Domain Generalization (In-the-Wild)

On the unconstrained In-the-Wild corpus ($N = 25{,}589$), the model records an optimal EER of 44.73% (ROC-AUC 0.5638). Polarity verification confirms that predictions are positively aligned with bona fide speech, but diverse recording hardware, reverberation, and unknown social media lossy compression create a profound acoustic distribution shift.

## V. Conclusion

We presented an acoustic forensics system for synthetic speech detection, tracing the progression from classical statistical baselines (11.46%–23.94% EER) to a deep multi-head self-attentive CRNN-LFCC countermeasure (6.70% EER, 0.2143 min-tDCF). While score-level stacked generalization was explored to fuse complementary feature spaces, empirical evaluation demonstrated that the stacker overfits clean development scores (10.74% Eval EER), confirming that standalone CRNN-LFCC offers superior operational generalization. Forensic stress-testing quantified model resilience under lossy codecs, exposed the critical impact of silence shortcut bias, and documented the domain shift on real-world audio. Future research will explore self-supervised front-ends (WavLM, HuBERT), acoustic room impulse response (RIR) augmentation, and multi-domain channel normalization.

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